Categories and Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Categories and Labels

> Prepare Categories and Labels: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Apply a **category** definition using explicit **evidence** and boundary rules

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-1/7, ai-2/8. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret an ordered sequence of sound-signal samples with a stated time interval

Reference example, not the only acceptable wording: Five samples, a 2 ms interval, and 8 ms from first to last. Consecutive pairs make gaps 0-2, 2-4, 4-6 and 6-8: four gaps between five samples.

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Look for: Execute a stated rule including its boundary

Reference example, not the only acceptable wording: At least 5 gives narrow, wide, wide. Greater than 5 gives narrow, narrow, wide. Only the equality case changes among these **input**s.

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an observed result from an explanation of how it was produced

Reference example, not the only acceptable wording: **Observation**s include that the door opened when the visitor approached and that its **label** says intelligent entrance.

A useful question asks whether a written sensor threshold, a model fitted on examples or a person controls opening. The **label** is an observed marketing claim, not **mechanism evidence**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

Concept to explain

Labels are human choices that may **overlap** or be unclear.

Term	Meaning
category	group created for a purpose
label	name assigned to an example
overlap	when more than one label can fit
ambiguity	when evidence allows multiple reasonable meanings

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This lesson establishes **label**s as task-dependent records produced under a guide. Use fictional event and equipment forms rather than categories about classmates.

The three categories deliberately separate known failures from unknown information, with known failure taking priority. A ready **label** checks a representation; it does not establish truth, quality, safety or permission.

Agreement is useful **evidence** about consistency, but shared mistakes can produce high agreement.

Readiness answer

The missing word is information the representation does not establish. Filling it without **evidence** adds an assumption; a later stage should preserve the uncertainty or request a check.

Paper calendar cards with the four worked field combinations and the printed policy

A second set of cards **label**led independently by partners before discussion

The equipment-request cases on a separate sheet for independent assessment; no accounts or personal data are required

End goal for the pictorial case

By the end, I can apply a shared **label**ling policy consistently, including missing information.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Categories and Labels using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT objects and a purpose for sorting

2. Process — the deciding step

PROCESS people define **label**s and assign examples

3. Output — the result

OUTPUT category names attached to items

Key idea

CORE IDEA **Label**s are human choices that may **overlap** or be unclear.

How the mechanism works

UNDER THE HOOD A **label** is an **operational definition** chosen for a task. Categories may be mutually exclusive, multi-**label**, hierarchical, subjective, or context-dependent.

Ambiguity in **label**s becomes **ambiguity** in **training** and evaluation.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A playground object can be both red and round, a toy and sports equipment, or safe in one situation and unsafe in another.

Step 1

STEP 1 — OBSERVE objects and a purpose for sorting

Step 2

STEP 2 — TRACE people define **label**s and assign examples

Step 3

STEP 3 — CONCLUDE **category** names attached to items

Model answer

MODEL REASONING Strong **label**s include a purpose and a rule for membership. **Overlap**ping objects may need multiple **label**s or a priority rule.

A revised definition should reduce **disagreement** without pretending every case is simple.

A claim to avoid

DO NOT OVERCLAIM **Label**s do not exist naturally in the data; people define them for a purpose.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Six cards show two circles, two triangles and two squares. Sort by shape.

Why can red and blue circles share a **label**?

Exact explanation to model

Why the result follows

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

Explain the added worked example

Explain the added worked example · 1

Two students are sorting fictional event cards for a classroom calendar. Their first instruction is Put the good cards together. Mina calls a colourful card good. Theo calls a plain card good because its room is easy to read.

They disagree even though both followed a reasonable interpretation of the word good.

Explain the added worked example · 2

A **category** is useful for a task when people can tell what **evidence** makes an example belong to it. An **operational definition** states the **observation**s or checks to perform.

For this paper exercise, we will check two printed fields: an event day and a room code. The task is to check the record against a classroom policy, not to judge the quality of the event or the person who wrote it.

Labels depend on the definition in use. A model fitted to **label**s can learn the distinctions those **label**s encode, including mistakes and inconsistencies.

Before asking a model to copy a **label**, ask whether people have a clear, useful guide for assigning it.

Explain the added worked example · 3

Our fictional calendar accepts a whole-number day from 1 through 30, including both endpoints, and exactly one room code, A or B. The month is already fixed for this exercise. A blank or unreadable field counts as unknown.

The categories are revise, review and ready. Use the checks in the order shown.

Explain the added worked example · 4

Known failure and unknown are different. Day 31 is known and outside the accepted range, so the card needs revision. An unreadable day might be valid or invalid; without another known failure, it needs review.

Never silently turn a blank into zero just to make a comparison possible.

Key terms

Term	Meaning
Category	A group defined by stated membership criteria
Operational definition	A definition precise enough to tell someone what to observe or do when assigning a category
Label	The category recorded for an example under a particular guide
Annotator	A person who assigns labels to examples
Disagreement	A case where annotators assign different labels to the same example

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Card fields	Relevant evidence	Label and reason
Day 12 room A	Both values satisfy the criteria	Ready because both checks pass
Day 31 room A	31 is outside 1 through 30	Revise because a known value fails
Day 12 room unreadable	No known failure but the room is unknown	Review until the room is checked
Day 31 room unreadable	The day already fails although the room is unknown	Revise because the first check has priority

For the last card, first read only what is known: day 31. Compare it with the allowed range and record a failure. The guide therefore selects revise before the missing-room check.

Checking the room later could reveal another problem, but it cannot make day 31 satisfy this policy.

A clear explanation cites the particular criterion. Saying It looks wrong is not enough. Write Day 31 exceeds the accepted maximum of 30 instead.

That explanation lets another person reproduce the check and challenge it if the recorded value or policy is different.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Categories and Labels

A fictional craft club **label**s cardboard-order cards ready, review or revise.

The facts we will use

> **Supplied facts**
>
> Allowed quantity: whole number 1–4. Required colour: red or blue.

> **Supplied facts**
>
> If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

AI ACADEMY / AI-1/7

Categories and Labels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read policy

Check known invalid fields before checking missing ones.



2. Card A

4 is allowed and red is allowed: ready.



3. Card B

5 is outside 1-4: revise, even though colour is missing.



4. Card C

2 is valid but colour is blank: review.

CHANGE ONE THING

Change B's quantity from 5 to 4; leave colour blank.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read policy	Check known invalid fields before checking missing ones.
2. Card A	4 is allowed and red is allowed: ready.
3. Card B	5 is outside 1–4: revise, even though colour is missing.
4. Card C	2 is valid but colour is blank: review.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure.

What this example does not establish

Keep the stated limit

Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Change one thing in the pictured case

Change one thing

Change: Change B’s quantity from 5 to 4; leave colour blank.

Predicted result

Prediction: B becomes review.

Why the result follows

Reason: The known failure is removed, but a required field is still missing.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Categories and Labels, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Six cards show two circles, two triangles and two squares. Sort by shape.

2. State the decision to explain

Write the question to explain: Why can red and blue circles share a **label**?

3. Reveal the worked reasoning

Reveal after **prediction**s: Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

Hold back the independent task

Keep for an independent attempt: For this toy sorter, label an object Seed-bearing when seeds are observed, Other when inspection establishes that it has no seeds, and Review when the needed contents cannot be observed. Sort an opened apple with visible seeds, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Day 0 room A is revise because 0 falls below the accepted minimum of 1; the known day needs correction.

Day 18 with a blank room is review because the known day passes and the room is unknown; the missing **evidence** must be checked. Two people **label**ling everything ready establishes agreement only.

They may share a mistake, ignore the criteria or work from inaccurate records. Correct field checks also do not confirm the event or its permission.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the missing-word example from Week 6. Ask what evidence a blank supplies and what it does not.

Stage	Teacher action
7 to 20 minutes	Give the vague good-card instruction and collect two different interpretations. Replace it with observable field criteria. Read the ordered checks aloud.
20 to 33 minutes	Trace all four worked cards. Emphasise the inclusive boundary and the priority of a known failure over an unknown field.
33 to 45 minutes	Pairs complete W1 and W2. Ask each learner to cite a criterion rather than a personal impression.
Session 2 start 0 to 8 minutes	Label two fresh calendar cards independently before comparing. Keep original labels visible during discussion.
8 to 23 minutes	Complete W3. Distinguish counting agreement from checking correctness against source fields and policy.
23 to 38 minutes	Collect W4 independently. Require four labels with reasons and a statement about the limit of ready.
38 to 45 minutes	Use W5 to demonstrate a policy-version conflict. Discuss the exit question and preview choosing useful input features.

Reduce help deliberately

Completion practice

Completion task: A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

Reasoning to listen for

Teacher check: Looks nice has no agreed observable criterion and can invite different preferences.

A usable replacement checks for a known day outside whole numbers 1 to 30 or a known room other than A/B and **label**s revise; otherwise any blank or unreadable required field gives review; otherwise **label** ready.

> Accept equivalent wording that preserves all criteria and priority.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Contrast unreadable day with day 0. One is unknown; the other is known to violate the stated range.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Explain why B originally was revise rather than review, then **label** its changed version.

Reasoned teacher answer

Quantity 5 violates the allowed range and failure has priority. With quantity 4 and missing colour, B is review.

Changed-case reasoning: B becomes review. The known failure is removed, but a required field is still missing.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Categories and Labels investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Design **label**s for playground objects; challenge ambiguous cards.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Every object must receive a confident **label**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A blank is the same as a failing value: Contrast unreadable day with day 0. One is unknown; the other is known to violate the stated range.

Review must win whenever any field is missing: Trace the guide in order for day 31 with a blank room. The known failure already selects revise under this policy.

Agreement proves the **label**s are true: Give two **annotator**s the same mistaken exclusive-boundary interpretation. Their agreement can coexist with incorrect **label**s.

A **category** describes the whole person or event: Return to the task scope. Ready checks two printed fields under one policy, not the worth of a person or the reality of an event.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Categories and Labels questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Both meet the circle definition. Colour does not change the agreed shape **label**. Apply one definition consistently.

Independent question

Independent question · withhold the key

For this toy sorter, label an object Seed-bearing when seeds are observed, Other when inspection establishes that it has no seeds, and Review when the needed contents cannot be observed. Sort an opened apple with visible seeds, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

Expected independent answer

Teacher key · Expected independent answer

The opened apple is Seed-bearing because its seeds are visible; the inspected pencil is Other because it has no seeds. The sealed box is Review: its contents have not been observed. These are **label**s for the toy task, not a biological definition of fruit.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Every object must receive a confident **label**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> One defensible rule is 'designed by its maker for repeated use', which excludes the disposable bottle. Another is 'used more than once', which includes it.
>
> State which task the **label**s serve, use one rule consistently and record unresolved cases; neither team should silently change definitions.

Explanations for the additional workbook examples

W1 Apply the calendar guide

Workbook question · W1

The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness.

Label these cards and give **evidence**: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

Teacher reasoning · W1

In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

W2 Replace a vague instruction

Workbook question · W2

A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

Teacher reasoning · W2

Looks nice has no agreed observable criterion and can invite different preferences.

A usable replacement checks for a known day outside whole numbers 1 to 30 or a known room other than A/B and **label**s revise; otherwise any blank or unreadable required field gives review; otherwise **label** ready.

Accept equivalent wording that preserves all criteria and priority.

W3 Count and investigate disagreement

Workbook question · W3

For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the **disagreement**s.

Can you decide which **annotator** is correct from these **label** lists alone? Explain what else you need.

Teacher reasoning · W3

Positions 1 and 3 agree, so there are two agreements out of four. Positions 2 and 4 disagree.

The lists alone do not show which **label**s are correct; inspect the corresponding card fields and the guide version, then compare each explanation with the criteria.

Agreement at positions 1 and 3 does not itself prove correctness.

W4 Transfer the method to a new record

Workbook question · W4

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

Teacher reasoning · W4

7 days X is ready: both pass, including the endpoint. 8 days Y is revise: duration exceeds 7.

Unreadable duration X is review: no known failure but duration is unknown. 8 days blank code is revise: the known duration failure has priority.

Ready only means the printed fields satisfy this policy; borrowing still requires the actual availability and authorisation **process**.

W5 Keep a policy change visible

Workbook question · W5

Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its **label** under each version.

Explain why mixing the two **label**s without recording the version could create a misleading **training** dataset.

Teacher reasoning · W5

Under Version 1 the card is revise; under Version 2 it is ready. Both can be correct relative to their different policies.

If the version is omitted, identical printed fields appear to have conflicting targets without explanation.

Record which policy produced each **label** and review or re**label** consistently for the intended task; do not erase the history.

Assess independent transfer

Independent question · withhold the key

Selected case: A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

Selected case answer

Teacher key · Selected case answer

7 days X is ready: both pass, including the endpoint. 8 days Y is revise: duration exceeds 7.

Unreadable duration X is review: no known failure but duration is unknown. 8 days blank code is revise: the known duration failure has priority.

Ready only means the printed fields satisfy this policy; borrowing still requires the actual availability and authorisation **process**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Categories and Labels to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 8: Features: Useful Clues. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a three-step checklist and highlight known versus unreadable fields using words as well as colour. Let learners explain orally while pointing to the criterion.

Extension: ask how a different priority policy would **label** mixed cases, and why that change must be documented rather than silently merged into the old **label**s.

W4 is secure when all four **label**s follow the new range and codes, the reasons cite the relevant **evidence**, the mixed failure/unknown case respects priority, and ready is not treated as authorisation.

If only the **label**s are correct, ask for the reasoning before recording mastery. After support, use new forms with a different maximum and codes for an uncoached retry.

Week 8 examines features: which recorded properties retain the distinctions a task needs, and which attractive-looking **input**s add little useful information.

Plan the delayed return

Suggested return points in this level: ai-1/8, ai-1/10, ai-1/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

TEACHER GUIDE · 7 OF 9

Prepare, teach and assess

> Use this preparation summary alongside the detailed Categories and Labels lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Apply a **category** definition using explicit **evidence** and boundary rules

Prepare the **label**led worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

Supported practice answer

Reasoned practice answer: Looks nice has no agreed observable criterion and can invite different preferences.

A usable replacement checks for a known day outside whole numbers 1 to 30 or a known room other than A/B and **label**s revise; otherwise any blank or unreadable required field gives review; otherwise **label** ready.

Accept equivalent wording that preserves all criteria and priority.

Check every learner before discussion: Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Contrast unreadable day with day 0. One is unknown; the other is known to violate the stated range.

Support without giving away the answer

Support: read the exact case aloud, enlarge the **label**led picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Change B's quantity from 5 to 4; leave colour blank. Require a **prediction** and a reason before comparing. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

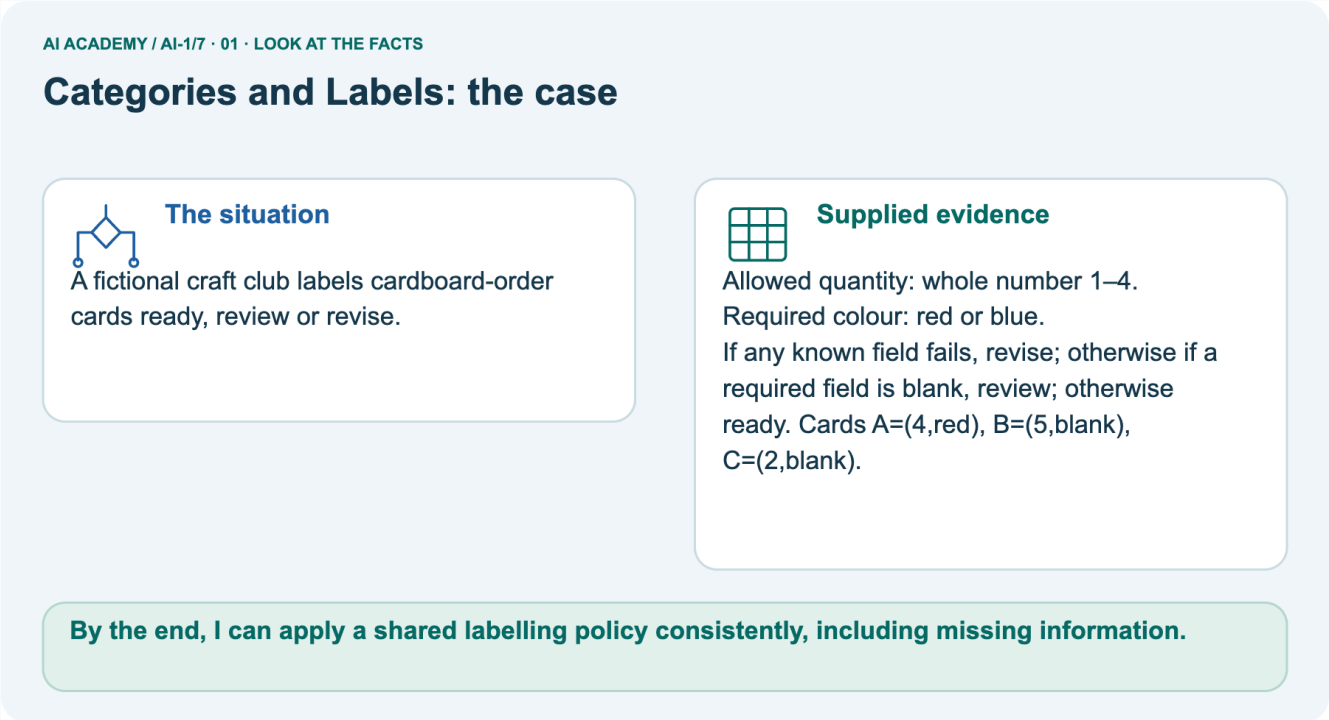
TEACHER GUIDE · 9 OF 9

Visual concept journey

> Show one Categories and Labels graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Categories and Labels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft club **label**s cardboard-order cards ready, review or revise. Supplied facts: Allowed quantity: whole number 1–4. Required colour: red or blue.

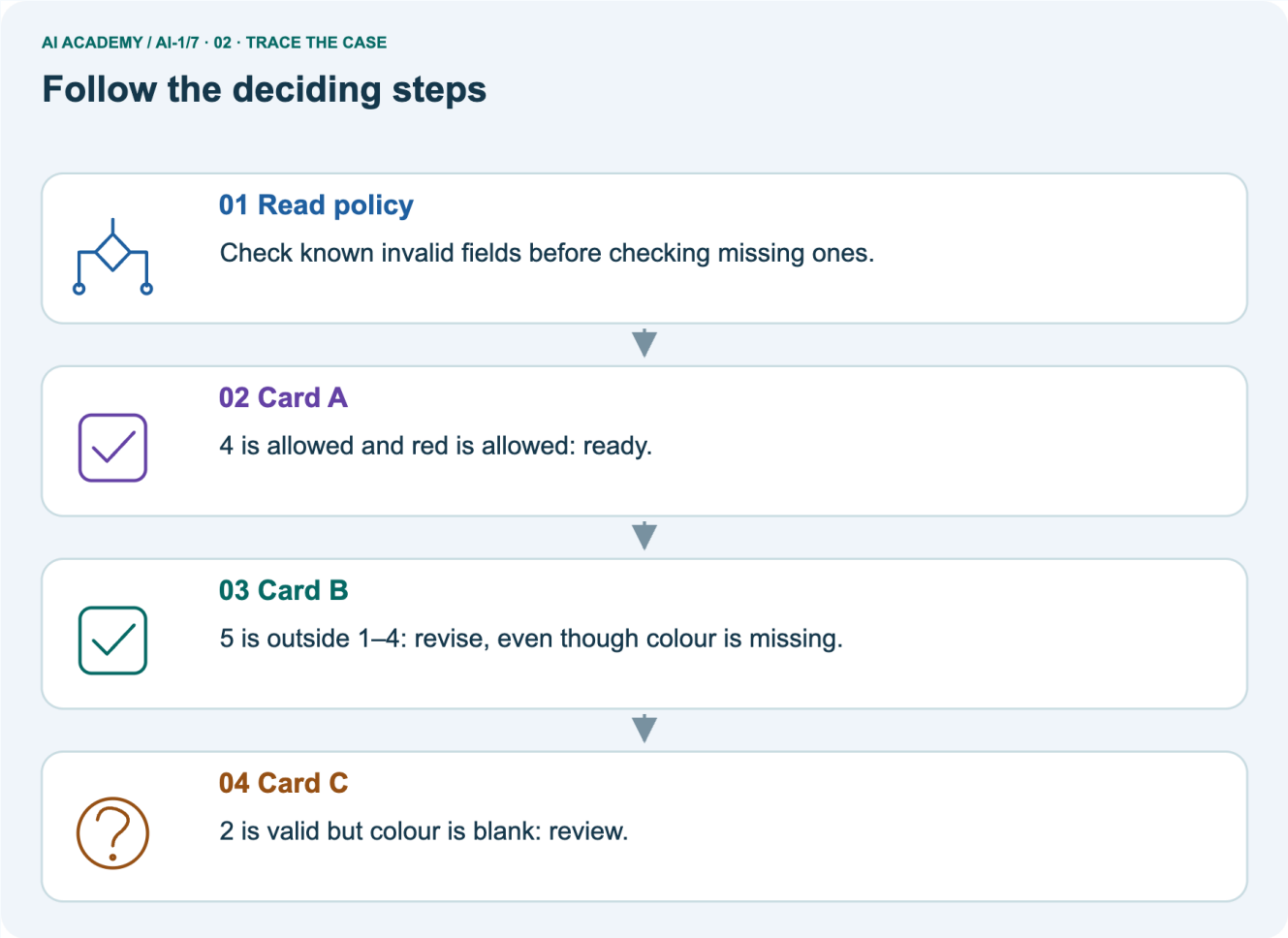
If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read policy: Check known invalid fields before checking missing ones. → Card A: 4 is allowed and red is allowed: ready. → Card B: 5 is outside 1–4: revise, even though colour is missing. → Card C: 2 is valid but colour is blank: review.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Three ordered checks for a calendar card



Follow the labelled picture with the matching explanation. Open the vector diagram

Three ordered checks for a calendar card. First check known values | Any known field fails its criterion | Revise even if another field is unknown.

Then check missing information | No known failure but a field is blank or unreadable | Review because **evidence** is incomplete.

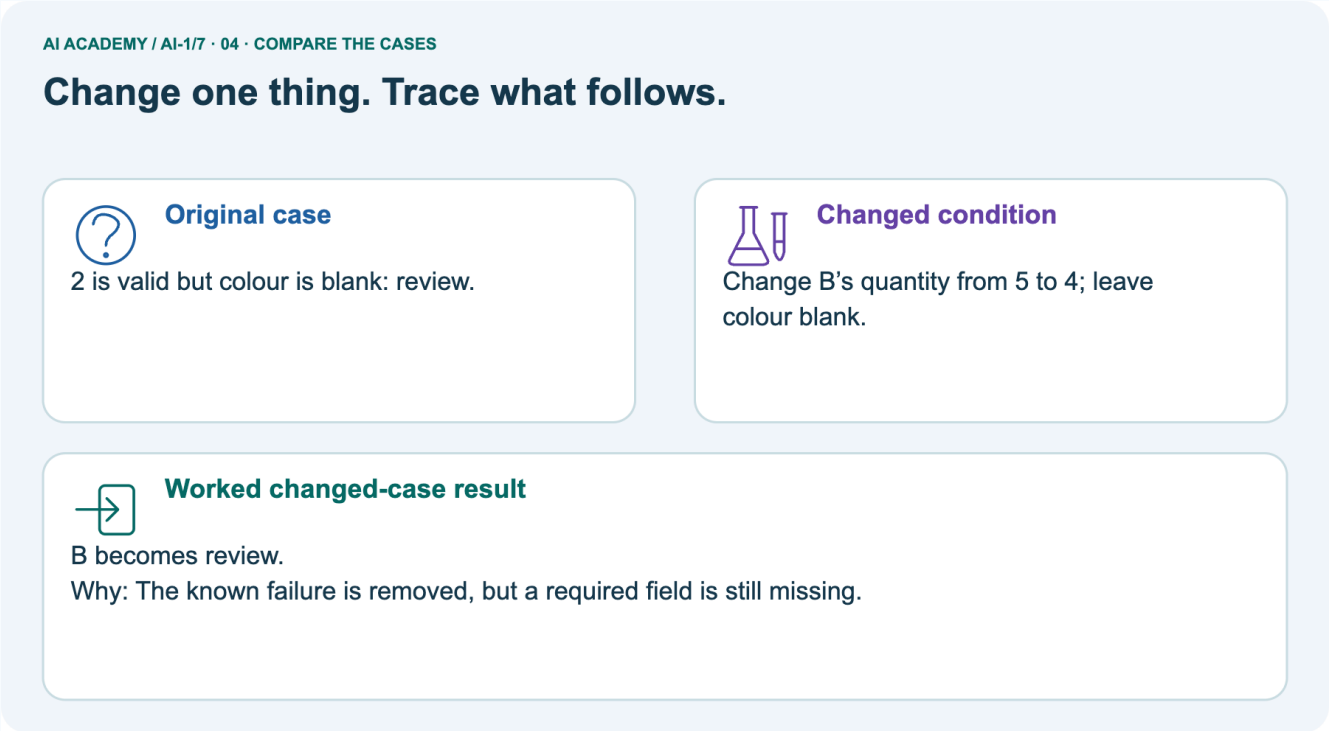
Otherwise all checks have passed | Day is 1 through 30 and room is A or B | Ready under this record policy

Read the **label**s and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the **label**s and values. Explain the relationship the model shows. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 2 is valid but colour is blank: review. Changed condition: Change B's quantity from 5 to 4; leave colour blank. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

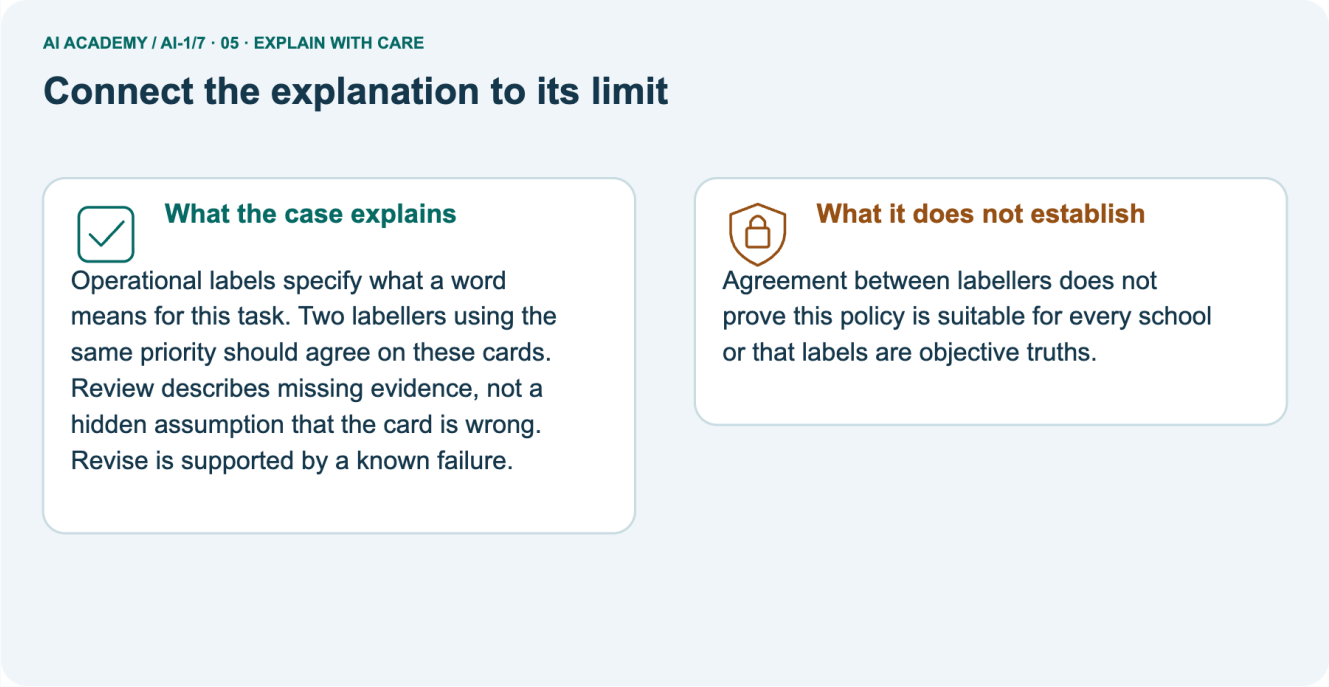
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one **label** or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The known failure is removed, but a required field is still missing.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure. Limit: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

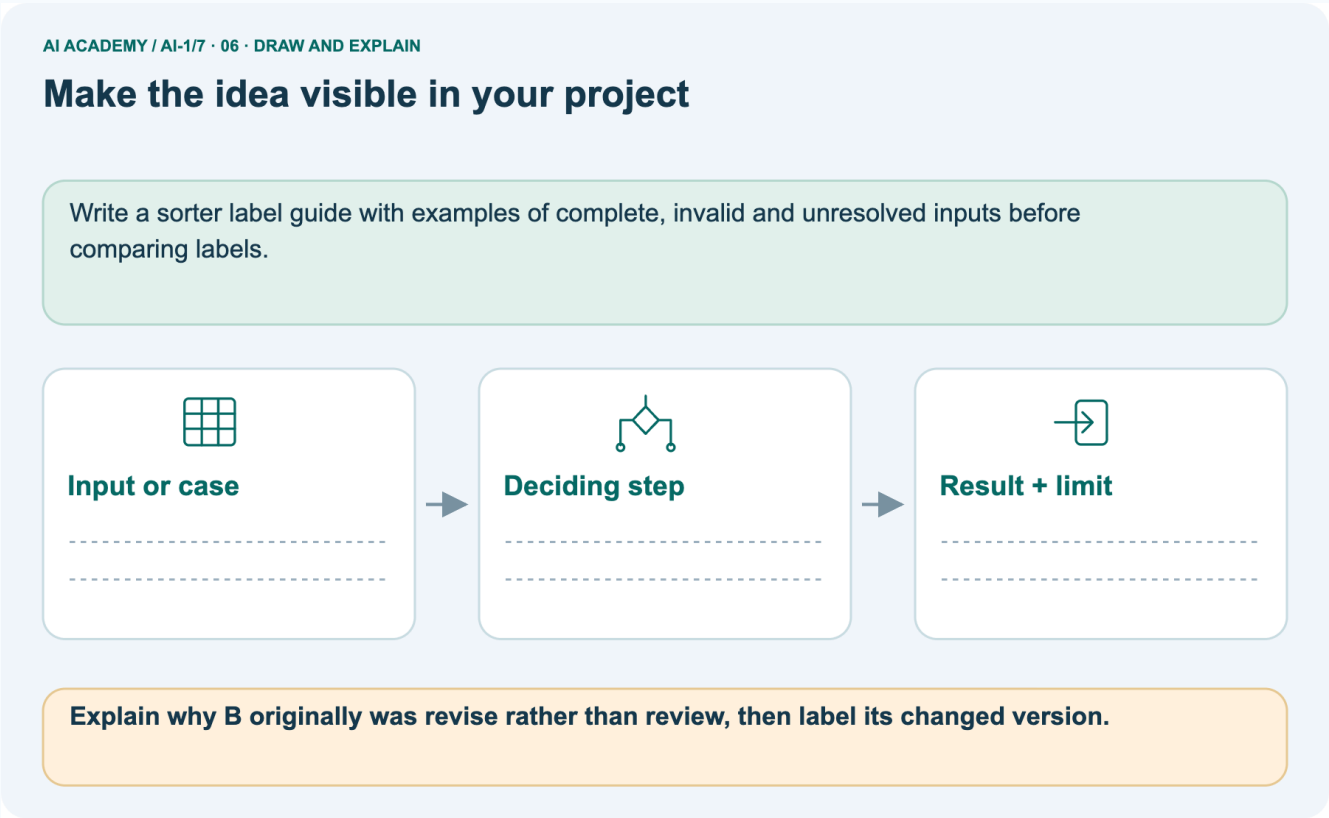
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B originally was revise rather than review, then **label** its changed version.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one **label** or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Quantity 5 violates the allowed range and failure has priority. With quantity 4 and missing colour, B is review.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.